MINTS: Testing Strict Motif-Detector Claims in Nucleotide Transformers

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Abstract

Genomic foundation models often predict biological labels well, but predictive success does not show that a model implements recognizable biological mechanisms. We study whether nucleotide transformers contain attention heads that can be justified as motif detectors under a strict mechanistic evidence standard. MINTS combines tokenization-aware QK/OV circuit analysis, residual-stream probing, probe controls, matched attention enrichment, activation patching, and sparse feature search on DNABERT-2 and a tested Nucleotide Transformer backend. DNABERT-2 layer-11 residual probes achieve AUROC values of 0.9137, 0.9383, 0.8954, and 0.8847 on promoter and splice-site tasks, showing strong label decodability. Controls sharpen the claim: random-label and position-only probes are near chance; splice-site probes remain about 0.25 AUROC above GC-only controls; promoter probes are close to strong GC-only baselines and should be interpreted cautiously. The strict CTCF test is negative: across 51,249 GM12878 CTCF sequences, no DNABERT-2 head meets the registered QK-to-motif criterion or matched attention-enrichment criterion. Batch activation patching finds promoter-TATA over-restoration and weaker splice-donor restoration, but OV readout and CTCF sparse-autoencoder audits remain weak. These results support a disciplined evidence stack for genomic mechanistic interpretability and show where representation-level success falls short of a strict motif-detector proof.

1 Introduction

Transformer models for nucleotide sequences now perform well on regulatory and splice-site prediction tasks [8, 4, 7]. However, the usual evidence for biological understanding remains predictive or post hoc: attribution maps, motif visualizations, or probes can identify useful correlations, but they do not by themselves explain how internal computations produce predictions [1, 13]. This distinction is central for mechanistic interpretability, where a claim about a component should survive direct tests of what it reads, where it attends, what it writes, and whether intervention on it changes the relevant behavior [5, 10, 6].

Nucleotide transformers are a useful setting for this question. Known motifs give natural ground-truth hypotheses, while tokenization and biological confounding make overinterpretation easy. A strong residual probe may exploit GC content, genomic-coordinate artifacts, or other distributional signals rather than a causal motif feature. Conversely, a negative circuit result can be informative: if no head passes pre-registered QK alignment and enrichment tests, then a high downstream AUROC should not be reported as evidence for a simple motif-local head.

This paper contributes MINTS, a reproducible pipeline and evidence standard for testing strict motif-detector claims in nucleotide transformers. We make four points. First, we give a tokenization-aware formulation of QK/OV analysis and activation patching for encoder-style genomic transformers.

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Second, we separate three levels of evidence that are often conflated: residual decodability, causal restoration under activation patching, and a strict motif-detector proof requiring QK alignment, motif-local enrichment, and causal effect. Third, we report a completed run on DNABERT-2 with a Nucleotide Transformer comparison backend, including probe controls designed to distinguish decodability from stronger causal-feature claims. Fourth, the main strict result is negative: under the registered CTCF criteria, neither backend yields a CTCF attention head that should be called a strict motif detector.

2 Related Work and Positioning

Mechanistic interpretability of transformers has developed increasingly precise tools for decomposing attention heads, locating causal pathways, and tracing computational graphs [5, 10, 2]. QK/OV decomposition is especially relevant here because it separates where a head attends from what it writes. Activation patching then gives an intervention-based test of whether the component matters for a behavior [6]. Recent work also makes the limits of current circuit tracing clear: even sophisticated attribution-graph studies explain only parts of the computation and remain expensive at long context lengths [9, 12].

Genomic deep-learning interpretability has a different tradition. Models such as DeepBind, SpliceAI, DNABERT, GPN, and Nucleotide Transformer show that neural sequence models learn useful biological regularities [1, 7, 8, 3, 4]. Many analyses in this area focus on motif visualization, attribution, saliency, or downstream probing. These are valuable, but they usually do not establish an internal algorithm. A saliency map may highlight a motif, and a probe may decode a label, while the model's actual computation may depend on a distributed feature, a correlated composition signal, or a different component entirely.

MINTS is positioned between these literatures. It does not claim to solve genomic interpretability in general, and it does not treat a high downstream score as mechanistic evidence. Instead, it asks a narrower question: when, if ever, is it justified to call a specific attention head a biological motif detector? This choice deliberately favors falsifiable criteria over narrative completeness. Negative results are therefore first-class evidence, because they identify where an appealing mechanistic story is not supported by the implemented tests.

3 Methods and Evidence Criteria

Transformer circuits. Let $\tau(x) = (t_1, \ldots, t_T)$ tokenize a nucleotide sequence into vocabulary items. The initial hidden state is

$$H^{(0)}_{j,:} = E_{\text{tok}}[t_j, :] + P_{j,:}. \tag{1}$$

For head h in layer l, with hidden state $H^{(l-1)}$, the attention pattern is

$$A^{(l)}_h = \text{softmax}\left(\frac{H^{(l-1)} W^Q_h (W^K_h)^T (H^{(l-1)})^T}{\sqrt{d_k}}\right), \tag{2}$$

and the head output is $A^{(l)}_h H^{(l-1)} W^V_h W^O_h$. Following transformer-circuit notation [5], the QK circuit $W^Q_h (W^K_h)^T$ controls attention scores, while the OV circuit $W^V_h W^O_h$ controls what is written to the residual stream. For nucleotide models, tokenization matters: a motif position may correspond to one nucleotide token, a fixed k-mer token, or a learned BPE span.

Strict motif-detector criterion. An attention head is treated as a candidate biological motif detector only when three signals converge. It must assign elevated attention mass to motif-support positions, its QK scores must rank motif-support tokens above matched background tokens, and patching its activation must restore the task-relevant scalar in motif-destroying counterfactuals. The registered thresholds used here are in Table 1. These thresholds intentionally make attention alone insufficient.

Activation patching. For a clean sequence x_c and motif-destroyed corrupted sequence x_{corr}, we patch a head output from the clean run into the corrupted run. The restoration metric is

$$\text{PM}(h, l) = \frac{f(x_{\text{patched}}) - f(x_{\text{corr}})}{f(x_c) - f(x_{\text{corr}})}, \tag{3}$$

Table 1: Registered evidence criteria for candidate motif circuits.

Claim	Statistic	Threshold
Motif localization	Attention enrichment ratio ρ_h	$\rho_h \geq 2.0$
Motif ranking	QK/motif Pearson correlation	$r \geq 0.5,\ p < 0.05$
Causal importance	Patching metric $\mathrm{PM}(h, l)$	$\mathrm{PM} \geq 0.5$
Strong causal importance	Same	$\mathrm{PM} \geq 0.75$
Residual decodability	Probe AUROC or macro-F_1	≥ 0.80 or $+0.10$
Robustness	Matched negative controls	No matched artifact

where f is a scalar target such as a probe decision function or class logit. $\mathrm{PM} = 0$ indicates no recovery, $\mathrm{PM} = 1$ indicates full recovery of the clean effect, and values above one indicate over-restoration rather than simple full recovery [6].

Residual probes and controls. Linear probes are trained on frozen mean-pooled residual vectors. Their role is diagnostic: a high probe score establishes linear decodability, not that the probe direction is causal or used by the model. We therefore compare residual probes with GC-content-only classifiers, position-only classifiers where coordinates are available, GC-matched test subsets, shuffled-label probes, and GC-content distribution shifts. This framing follows linear-representation work while preserving the weaker interpretation that probes support [11].

4 Experimental Setup

The primary model is `zhihan1996/DNABERT-2-117M`, an encoder-style genomic transformer with learned BPE tokenization. The comparison backend is `InstaDeepAI/nucleotide-transformer-v2-100m-multi-species`, used as a tested fixed-6mer Nucleotide Transformer checkpoint [4]. The run exports DNABERT-2 QK and OV matrices for all 12 layers and 12 heads per layer, caches residual vectors for layers 0, 5, and 11, and uses custom Hugging Face forward hooks for attention-output patching because the installed TransformerLens path does not expose all required DNABERT-2 per-head tensors.

The residual-probe tasks are TATA promoter, non-TATA promoter, splice donor, and splice acceptor classification. The configured splits contain 5,062/212 train/test examples for TATA promoters, 30,000/1,372 for non-TATA promoters, and 30,000/3,000 for each splice task. The strict CTCF scan uses 51,249 prepared GM12878 CTCF peak sequences and the JASPAR MA0139.1 CTCF motif, aligned to exact overlapping DNABERT-2 BPE token spans. Batch patching uses motif-destroying clean/corrupted pairs for promoter-TATA and splice-donor tasks.

The pipeline writes small review-facing CSV, JSON, and figure artifacts while treating large activation, circuit-matrix, token-score, and sparse-autoencoder files as reproducible generated outputs. The full run takes about eight hours on the tested CUDA setup. The most expensive stages are the cross-model comparison and strict all-layer CTCF scans, so the appendix includes both full reproduction commands and a reduced smoke-run command for checking the code path on a smaller subset.

5 Results

Residual decodability is strong but not uniformly clean. DNABERT-2 layer-11 residual probes exceed the registered residual-decodability threshold for all four tasks. Table 2 combines headline AUROC values with the most important controls. Random-label probes collapse to chance, and position-only metadata does not explain the results. The splice-site tasks survive stronger checks: donor and acceptor residual probes remain about 0.25 AUROC above GC-only baselines on matched subsets and remain strong under GC shifts. The promoter tasks require caution because GC-only baselines are already high; their residual-probe advantage over GC-only controls is small on the matched comparison.

This distinction changes the interpretation of the headline probe table. The splice-site result is evidence that DNABERT-2's frozen residual stream contains linearly accessible information beyond simple GC composition or coordinate metadata. The promoter result is weaker mechanistically: it

Table 2: DNABERT-2 layer-11 probe-control summary. Numeric columns report AUROC; random-label values are means over eight shuffled-label runs.

Task	Resid.	GC-only	Pos.-only	GC-match	Rand.	GC-shift range
promoter_tata	0.9137	0.8956	0.4136	0.9137	0.4700	0.6182–0.8030
promoter_no_tata	0.9383	0.9088	0.3865	0.9383	0.5129	0.6312–0.8368
splice_sites_donors	0.8954	0.6560	0.4414	0.8944	0.5064	0.8432–0.8669
splice_sites_acceptors	0.8847	0.6361	0.4461	0.8838	0.5005	0.8573–0.8580

Table 3: Strict mechanistic evidence summary from the completed run.

Evidence channel	Executed test	Best observed result	Met?
Residual decodability	Layer-11 probes on four benchmark tasks	AUROC range 0.8847–0.9383	Yes
Probe controls	GC, position, matched, random-label, and shift controls	Splice robust; promoter close to strong GC baselines	Mixed
CTCF QK alignment	144 DNABERT-2 heads vs. BPE-correct CTCF scores	Best $r = 0.3004$	No
CTCF enrichment	Matched motif/background attention mass	Best $\rho_h = 1.3130$	No
Promoter-TATA patching	327 batch denoising pairs	Best mean $\mathrm{PM} = 1.4029$	Yes
Splice-donor patching	500 batch denoising pairs	Best mean $\mathrm{PM} = 0.5485$	Partial
OV readout audit	TATA probe direction vs. OV singular vectors	Top cosine 0.1261; self-gain -0.0326	Weak
CTCF SAE search	Residual and MLP sparse features	Best motif cosine 0.1158	Weak
Strict CTCF detector	QK, enrichment, and causal evidence in one head	No passing QK/enrichment candidate	No

remains a valid decodability result, but the controls show that promoter-associated composition can explain much of the signal. This is why the paper avoids claiming that the promoter probes have isolated a clean causal promoter motif direction.

The strict CTCF motif-detector test fails. The CTCF scan produced 281,915 motif-support tokens and 1,843,874 finite token positions after BPE span correction. Across all 144 DNABERT-2 heads, the best QK-to-motif correlation is $r = 0.3004$, below the registered $r \geq 0.5$ criterion. The best matched attention-enrichment ratio is $\rho_h = 1.3130$, below the registered $\rho_h \geq 2.0$ criterion. The tested Nucleotide Transformer backend also fails to recover a strict CTCF candidate, with best $r = 0.0192$ and best $\rho_h = 1.00009$. Thus, changing to the tested fixed-6mer backend does not rescue the CTCF motif-local head hypothesis.

The negative CTCF result is not a failure of the pipeline; it is the point of using a registered evidence stack. With more than 1.8 million finite token positions, small correlations can be statistically nonzero while remaining too small to support a mechanistic claim. The all-layer enrichment scan similarly prevents a common attention-overinterpretation mistake: a head can have visible structure without assigning enough mass to matched motif-support positions to count as a motif-local detector.

Patching gives task-specific causal signals, not a full circuit. Batch promoter-TATA patching over 327 usable pairs identifies layer 4 head 8 with mean $\mathrm{PM} = 1.4029$; layer 6 head 8 and layer 2 head 7 are also high. Because $\mathrm{PM} > 1$ overshoots the clean-minus-corrupted effect, this is a methodologically sensitive causal signal rather than a simple full-recovery story. Batch splice-donor patching over 500 pairs identifies layer 1 head 8 with mean $\mathrm{PM} = 0.5485$, a weaker but threshold-crossing effect. These interventions support task-relevant internal computation, but they do not identify a strict CTCF motif detector because the CTCF QK and enrichment screens have no passing head.

OV, SAE, and cross-model audits remain conservative. The TATA OV readout audit for layer 2 head 7 finds only weak alignment with the trained TATA probe direction: the top output-write singular-vector cosine is 0.1261 and the probe self-gain through the OV matrix is -0.0326. Sparse-autoencoder CTCF feature search is also weak, with best residual and MLP motif cosines of 0.0884 and 0.1158. In the protocol-level cross-model comparison, DNABERT-2 residual probes outperform the tested Nucleotide Transformer backend on the same tasks, but this is a checkpoint- and protocol-specific result, not a general claim about tokenization families or all Nucleotide Transformer variants.

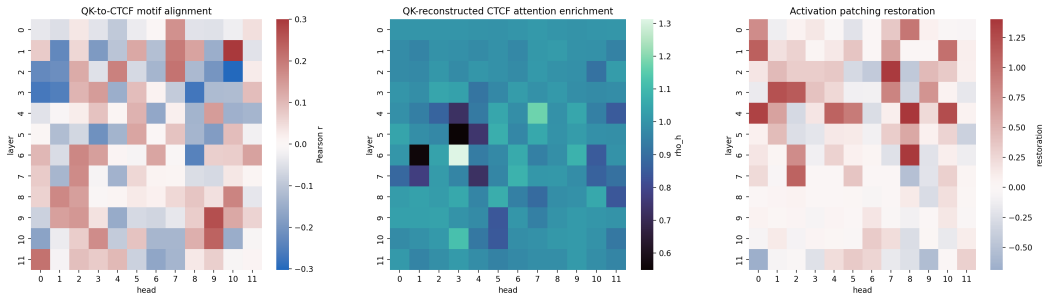

Figure 1: Mechanistic evidence heatmaps. Left: DNABERT-2 CTCF QK-to-motif correlations. Middle: matched attention-enrichment ratios. Right: promoter-TATA batch patching. The CTCF heatmaps show a negative strict-proof result, while the TATA patching heatmap shows task-specific causal over-restoration.

6 Limitations

This study is intentionally scoped to small and medium genomic transformers and does not claim exhaustive explanation. Current large-model circuit-tracing work also emphasizes partial recovery of computation rather than complete mechanistic accounts [2, 9]. The strict CTCF failure applies to the selected models, CTCF intervals, motif scorer, BPE span alignment, and registered thresholds; it is not an impossibility theorem for all models or motifs.

Linear probes remain limited evidence. They can reveal decodable information while exploiting correlated sequence composition, split artifacts, or metadata. The added GC, position, matched, random-label, and shift controls reduce this risk, especially for splice-site tasks, but promoter decodability remains entangled with GC composition. Activation patching is causal but incomplete: head-level restoration does not characterize the full circuit, and over-restoration can reflect denominator sensitivity, probe geometry, or out-of-distribution patched states. Finally, biological annotations are incomplete, and a single JASPAR PWM cannot represent all relevant regulatory mechanisms.

7 Conclusion

MINTS turns a common interpretability temptation into a falsifiable test: before calling an attention head a biological motif detector, require aligned QK scores, motif-local attention enrichment, and causal restoration. On DNABERT-2, promoter and splice-site labels are strongly linearly decodable from late residual states, and splice-site decodability survives simple composition and position controls. But the strict CTCF motif-detector proof fails across all tested heads, and the task-specific patching effects do not change that conclusion. The contribution is therefore both positive and negative: a reproducible evidence stack for nucleotide-transformer interpretability, and a concrete demonstration that high representation-level performance should not be upgraded into a circuit-level biological claim without stronger evidence.

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A Theoretical Details

Definition 1 (Biological Motif Detector)**.** *An attention head h in layer l functions as a biological motif detector for motif m if there exists a nonempty set of relevant query positions $\mathcal{Q}_m$ and a threshold τ such that, for each $i \in \mathcal{Q}_m$,*

$$\mathbb{E}\left[\sum_{j \in \mathrm{supp}_m(x)} A_h^{(l)}[i,j] \mid \mathrm{supp}_m(x) \neq \varnothing\right] \geq \tau \tag{4}$$

and the corresponding expectation conditioned on $\mathrm{supp}_m(x) = \varnothing$ is below τ. Here $\mathrm{supp}_m(x)$ is the set of key positions whose local sequence windows instantiate motif m.

Theorem 1 (Layer-1 Motif Detection via QK Circuits)**.** *Consider a layer-1 attention head operating on $H^{(0)}$. Let $s_m(j)$ denote a scalar motif score associated with key position j. If for a fixed query position i there are constants $a_i \in \mathbb{R}$ and $\gamma > 0$ such that*

$$q_i^T k_j = a_i + \gamma s_m(j) \tag{5}$$

for all admissible key positions j, then the induced attention distribution is monotonically increasing in $s_m(j)$. If motif-bearing positions are separated from background positions by a positive margin in s_m, the head is a motif detector for m.

Proof. The attention logit is $\ell_{ij} = q_i^T k_j/\sqrt{d_k} = a_i/\sqrt{d_k} + (\gamma/\sqrt{d_k})s_m(j)$. Since $\gamma/\sqrt{d_k} > 0$, scaling preserves the ordering of motif scores. The exponential and softmax are strictly monotone in each logit when other logits are fixed, so positions with larger motif scores receive larger attention weights. A positive separation margin therefore induces an attention-mass threshold that separates motif-bearing from background positions. □

Lemma 1 (Approximate Extension Under Linear Decodability)**.** *Suppose that at layer $l-1$ there exists a vector u_m such that*

$$u_m^T h_j^{(l-1)} = s_m(j) + \epsilon_j, \qquad |\epsilon_j| \leq \delta. \tag{6}$$

If a head's QK score is affine in this decodable direction and the clean motif/background score margin is greater than 2δ, then the attention ordering between motif and background positions is preserved.

Proof. For motif-bearing j_m and background j_b, the score difference is proportional to $s_m(j_m) - s_m(j_b) + \epsilon_{j_m} - \epsilon_{j_b}$. The perturbation term lies in $[-2\delta, 2\delta]$. A clean separation greater than 2δ keeps the sign of the score difference fixed, so softmax preserves the motif/background ordering. □

B Implementation and Reproducibility Details

The completed run started at 13:27:07 local time on April 14, 2026 and ended at 21:26:30 local time the same day. The manifest reports 28,760.213 seconds, or 7.989 hours. The longest steps were cross-model tokenization comparison, strict mechanistic proofs, and systematic causal intervention. The inspected result tree contains 125 files totaling about 4.71 GB, including CSV/JSON summaries, figures, NPZ archives, and sparse-autoencoder checkpoints.

The anonymized project repository is available at `https://anonymous.4open.science/r/MINTS/`. It contains the project code, configuration, local paper source, and reproduction instructions. The repository currently serves as the single source of truth for the submitted code; an explicit repository license has not yet been added.

The main reproduction commands are:

```
git clone https://anonymous.4open.science/r/MINTS/ MINTS
cd MINTS
python -m pip install -r requirements.txt
python main.py
python main.py --only-probe-controls
```

A smaller smoke run is:

```
python main.py --max-probe-train 512 --max-probe-test 256
  --max-qk-alignment-sequences 128
  --max-cross-model-qk-alignment-sequences 128
  --max-feature-search-sequences 128 --sae-epochs 1
```

Important verification artifacts include:

- `results/pipeline_run.json`
- `results/tables/linear_probe_metrics.csv`
- `results/tables/linear_probe_controls.csv`
- `results/qk_alignment/ctcf_qk_alignment.csv`
- `results/enrichment/ctcf_qk_alignment_matched_attention_enrichment.csv`
- `results/patching/promoter_tata_batch_dnabert_activation_patching.csv`
- `results/patching/splice_sites_donors_batch_dnabert_activation_patching.csv`

The heatmaps used in the paper are generated under `results/figures/`.

C Additional Results

The full configured-split residual probe AUROCs are 0.9137 for TATA promoters, 0.9383 for non-TATA promoters, 0.8954 for splice donors, and 0.8847 for splice acceptors. The corresponding AUPRC values are 0.9241, 0.9475, 0.9049, and 0.8954.

For promoter-TATA batch patching, 327 token-shape-preserving pairs were usable with no near-zero denominator failures. The best mean score was layer 4 head 8 with $\mathrm{PM} = 1.4029$, followed by layer 6 head 8 with $\mathrm{PM} = 1.3791$ and layer 2 head 7 with $\mathrm{PM} = 1.3642$. For splice donors, 500 pairs were usable with no denominator failures, and the strongest mean restoration was layer 1 head 8 with $\mathrm{PM} = 0.5485$.

For the tested Nucleotide Transformer backend, residual-probe AUROCs were 0.6502, 0.6976, 0.5695, and 0.5647 on the same four tasks. DNABERT-2 therefore outperformed this backend by 0.2408–0.3259 AUROC in this protocol. This comparison should not be read as a general ranking of tokenizers or genomic transformer families.

D Declaration of LLM Usage

LLM-assisted tools were used during the project for mathematical exposition refinement, debugging support, and code-generation drafts through Codex with state-of-the-art GPT-5.4/GPT-5.5-class models. The research direction, core problem decomposition, step-by-step implementation decisions, and final scientific interpretation were directed by the author. The author reviewed the generated code, checked it for errors and failure modes, corrected mistakes, and takes responsibility for the final manuscript, code, and claims.

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   Answer: [Yes]

   Justification: The abstract and introduction state the scope as an evidence stack for nucleotide-transformer interpretability, and the results distinguish residual decodability, causal patching, and the negative strict CTCF motif-detector test.

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- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Answer: [Yes]

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